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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2015 Half-Yearly Examination

FORM VI

MATHEMATICS EXTENSION 1

Tuesday 24th February 2015

General Instructions

- Writing time — 1 hour and 30 minutes
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 55 Marks

- All questions may be attempted.

Section I – 7 Marks

- Questions 1–7 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 48 Marks

- Questions 8–11 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Staple your answers in a single bundle.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eight.
- Write your candidate number on this question paper and hand it in with your answers.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 114 boys

Examiner

LYL

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

1

A parabola has a vertex at $(0, 4)$, focal length 2 units and its axis is parallel to one of the coordinate axes. Which equation fits the given conditions?

(A) $(y + 4)^2 = -4x$

(B) $(y - 4)^2 = 4x$

(C) $x^2 = -8(y - 4)$

(D) $x^2 = 8(y + 4)$

QUESTION TWO

1

What is the period of $y = -3 \sin \frac{1}{2}x$?

(A) $\frac{\pi}{2}$

(B) 2π

(C) π

(D) 4π

QUESTION THREE

1

If θ is the acute angle between the lines $y = -\frac{1}{3}x - 3$ and $y = 2x + 3$, then the value of $\tan \theta$ is:

(A) 7

(B) 1

(C) -7

(D) -1

Exam continues next page ...

QUESTION FOUR

1

What is the Cartesian equation of the curve $x = at^2$, $y = 2at$?

(A) $y^2 = 4ax$

(B) $y^2 = 2ax$

(C) $x^2 = 4ay$

(D) $x^2 = 2ay$

QUESTION FIVE

1

The angle θ satisfies $\cos \theta = \frac{4}{5}$ and $-\frac{\pi}{2} < \theta < 0$. What is the value of $\sin 2\theta$?

(A) $\frac{24}{25}$

(B) $-\frac{24}{25}$

(C) $\frac{7}{25}$

(D) $-\frac{7}{25}$

QUESTION SIX

1

What is the derivative of $\tan^{-1} \frac{1}{x}$?

(A) $\frac{1}{1+x^2}$

(B) $-\frac{1}{x^2+1}$

(C) $-\frac{x^2}{x^2+1}$

(D) $\frac{x^2}{x^2+1}$

QUESTION SEVEN

1

What is the domain and range of $y = 4 \cos^{-1} 3x$?

- (A) The domain is $-\frac{1}{3} \leq x \leq \frac{1}{3}$ and the range is $-2\pi \leq y \leq 2\pi$.
- (B) The domain is $-3 \leq x \leq 3$ and the range is $-2\pi \leq y \leq 2\pi$.
- (C) The domain is $-\frac{1}{3} \leq x \leq \frac{1}{3}$ and the range is $0 \leq y \leq 4\pi$.
- (D) The domain is $-3 \leq x \leq 3$ and the range is $-2\pi \leq y \leq 2\pi$.

————— End of Section I —————

Exam continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION EIGHT (12 marks) Use a separate writing booklet. **Marks**

(a) A sector has arc length 6 units and radius 4 units. Find the exact area of the sector. **2**

(b) Write down the exact value of:

(i) $\cos^{-1} \left(\frac{1}{2} \right)$ **1**

(ii) $\tan \frac{5\pi}{3}$ **1**

(c) Differentiate the following:

(i) $\tan \frac{x}{3}$ **1**

(ii) $e^x \sin x$ **2**

(iii) $\cos^3 x$ **1**

(d) Find:

(i) $\int \sec^2 \frac{x}{3} dx$ **1**

(ii) $\int \frac{4}{25 + x^2} dx$ **1**

(e) Find the exact value of $\cos \left(\sin^{-1} \frac{1}{3} \right)$. **2**

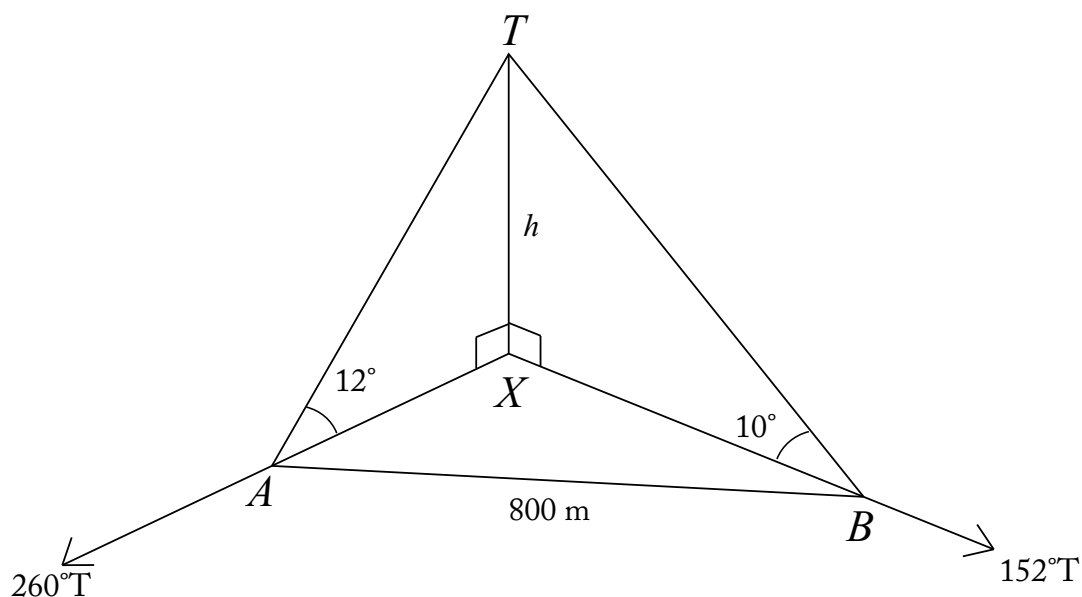
QUESTION NINE (12 marks) Use a separate writing booklet.

Marks

- (a) The radius r of a circle is increasing such that the rate of increase of the area of the circle is $\pi^2 r \text{ cm}^2/\text{s}$. Calculate the rate of increase of the radius. **2**
- (b) Consider the function defined by $f(x) = x^2 - 4$, for $x \leq 0$.
- (i) Draw a neat sketch of the function $y = f(x)$, for $x \leq 0$, clearly showing any intercepts with the axes. **1**
- (ii) Sketch the graph of the inverse function $y = f^{-1}(x)$. **1**
- (iii) State the domain of the inverse function $y = f^{-1}(x)$. **1**
- (c) Write down the general solution of $\cos x = \frac{\sqrt{3}}{2}$. Leave your answer in radians. **2**
- (d) Evaluate $\int_0^{\frac{\pi}{2}} \cos^2 x \, dx$. **2**
- (e) Use mathematical induction to prove $13^n - 1$ is divisible by 3 for all positive integers n . **3**

QUESTION TEN (12 marks) Use a separate writing booklet.**Marks**(a) Evaluate $\lim_{x \rightarrow 0} \frac{x}{\tan 3x}$. You must show working. 1(b) Find the equation of the normal to $x^2 = 12y$ at the point $(6p, 3p^2)$. Leave your answer in general form. 2(c) Evaluate $\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{dx}{\sqrt{1-2x^2}}$. 2(d) Solve the equation $3 \tan 2\theta = 2 \tan \theta$, for $0 \leq \theta \leq 2\pi$. 3

(e)



In diagram above TX represents a vertical tower of height h metres standing on the horizontal plane AXB . Two men 800 metres apart on the same plane observe the top of the tower. One man at point A is on a bearing of 260° T from the tower and the angle of elevation to the top of the tower is 12° . The second man at point B is on a bearing of 152° T from the tower and the angle of elevation to the top of the tower is 10° .

(i) Using a diagram, or otherwise, explain why $\angle AXB = 108^\circ$. 1(ii) Express AX in terms of h . 1(iii) Find the height of the tower to the nearest metre. 2

QUESTION ELEVEN (12 marks) Use a separate writing booklet.**Marks**

- (a) Express $8 \cos x + 15 \sin x$ in the form $R \cos(x - \phi)$ where $R > 0$ and $0^\circ < \phi < 360^\circ$.
In your answer, give the angle ϕ correct to the nearest degree.

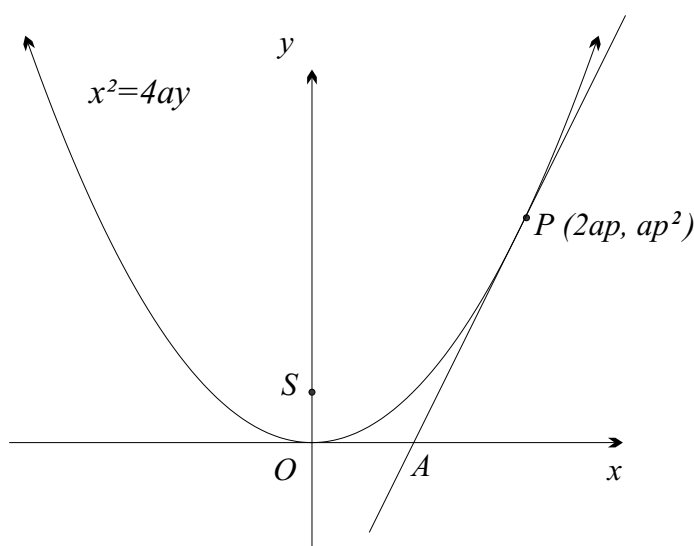
2

- (b) Prove the identity below using the substitution $t = \tan \frac{\theta}{2}$.

2

$$\frac{\sin \theta - 1 + \cos \theta}{\sin \theta + 1 - \cos \theta} = \frac{1 - \tan \frac{\theta}{2}}{1 + \tan \frac{\theta}{2}}$$

- (c)



In the diagram above the tangent to the parabola $x^2 = 4ay$ at the point $P(2ap, ap^2)$ meets the tangent to the vertex at the point A . The equation of the tangent at P is $y = px - ap^2$. (Do not prove this.)

- (i) If S is the focus, prove that SA is perpendicular to PA .
- (ii) It is given that R is the centre of a circle which passes through P , S and A . Determine the equation of the locus of R as P varies.
- (d) Consider the curves $y = \sin x$ and $y = \sin^2 x$, where $0 \leq x \leq \frac{\pi}{2}$.
- (i) Explain why $\sin^2 x \leq \sin x$, for $0 \leq x \leq \frac{\pi}{2}$.
- (ii) Find the volume of revolution generated when the area between the two curves is rotated about the x -axis.

2**2****1****3**

End of Section II

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Section 1 Q1 C, Q2 D, Q3 A, Q4 A, Q5 B, Q6 B, Q7 C.

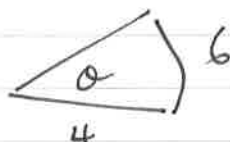
Section 2

Q8 a)

$$l = r\theta$$

$$6 = 4\theta$$

$$\theta = \frac{3}{2} \text{ radians } \checkmark$$



$$A = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times 4^2 \times \frac{3}{2}$$

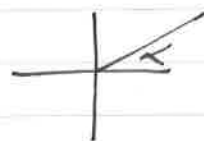
$$= 12 \text{ units}^2 \checkmark$$

b) i) let $\alpha = \cos^{-1} \frac{1}{2}$ where $0 < \alpha \leq \pi$

$$\cos \alpha = \frac{1}{2}$$

α in 1st quad

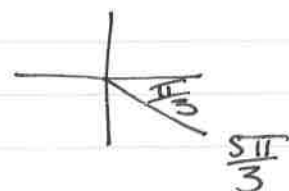
$$\alpha = \frac{\pi}{3} \checkmark$$



ii) let $\alpha = \tan^{-1} \frac{\pi}{3}$

$$= -\tan^{-1} \frac{\pi}{3}$$

$$= -\sqrt{3} \checkmark$$



c) i) $y = \tan \frac{x}{3}$
 $y' = \frac{1}{3} \sec^2 \left(\frac{x}{3} \right) \checkmark$

ii) $y = e^x \sin x$

$$y = uv$$

$$y' = v u' + u v'$$

$$= e^x \sin x + e^x \cos x$$

$$= e^x (\sin x + \cos x) \checkmark$$

$$u = e^x$$

$$v = \sin x$$

$$u' = e^x$$

$$v' = \cos x$$

iii) $y = \cos^3 x$

$$y' = -3 \cos^2 x \sin x \checkmark$$

d) i) $\int \sec^2 \frac{x}{3} dx = 3 \tan \frac{x}{3} + c \checkmark$

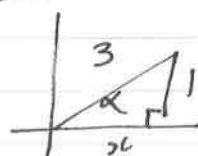
ii) $\int \frac{4}{25+x^2} dx = \frac{4}{5} \tan^{-1} \frac{x}{5} + c \checkmark$

e) let $\alpha = \sin^{-1} \frac{1}{3}$

$$\sin \alpha = \frac{1}{3}$$

$$\cos \alpha = \frac{2\sqrt{2}}{3} \checkmark$$

$$-\frac{\pi}{2} < \alpha \leq \frac{\pi}{2}$$



$$x^2 + 1^2 = 3^2$$

$$x = 2\sqrt{2} \checkmark$$

Q 9 a) $A = \pi r^2$

$$\frac{dA}{dr} = 2\pi r$$

$$\frac{dA}{dt} = \pi^2 r$$

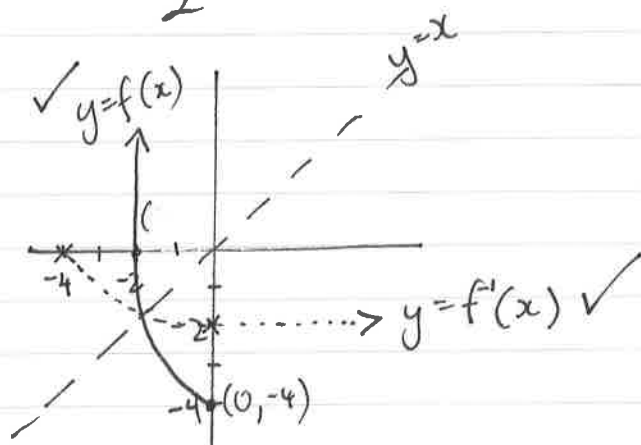
$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$\pi^2 r = 2\pi r \times \frac{dr}{dt} \quad \checkmark$$

$$\frac{dr}{dt} = \frac{\pi^2 r}{2\pi r}$$

$$= \frac{\pi}{2} \text{ cm/s} \quad \checkmark \text{ units required.}$$

b)



$$f(x) = x^2 - 4 \quad x \leq 0$$

Domain of $y = f^{-1}(x)$ $x \geq -4$ $\checkmark$

c) $\cos x = \frac{\sqrt{3}}{2}$

$$x = \cos^{-1} \frac{\sqrt{3}}{2} + 2n\pi \quad \text{or} \quad -\cos^{-1} \frac{\sqrt{3}}{2} + 2n\pi \quad n \in \mathbb{Z}$$

$$= 2n\pi \pm \frac{\pi}{6} \quad n \in \mathbb{Z}$$

$\checkmark \checkmark$ one mark for $\frac{\pi}{6}$

d) $\int_0^{\frac{\pi}{2}} \cos^2 x \, dx$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2} (1 + \cos 2x) \, dx$$

$$= \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}} \quad \checkmark$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{2} + \frac{1}{2} \sin \pi \right) - (0 + 0) \right]$$

$$= \frac{\pi}{4} \quad \checkmark$$

Q9e) Prove $13^n - 1$ div. by 3 for all positive integers n

A: When $n=1$

$$13^1 - 1 = 12 \text{ which is div. by } 3$$

$\therefore$ Statement true for $n=1$ ✓

B: Assume the statement holds true for $n=k$
where k is a positive integer.

$$\text{ie. } 13^k - 1 = 3M \text{ where } M \text{ is a positive integer.}$$

$$13^k = 3M + 1 \quad *$$

Must prove true for $n=k+1$

$$\text{ie. } 13^{k+1} - 1 = 3N \text{ where } N \text{ is a positive integer.}$$

$$\text{LHS} = 13^{k+1} - 1 \quad \checkmark$$

$$= 13^k \cdot 13^1 - 1$$

$$= (3M + 1)13 - 1$$

$$= 13 \times 3M + 13 - 1$$

$$= 13 \times 3M + 12$$

$$= 3(13M + 4) \quad \checkmark$$

using the induction hypothesis

C. It follows from parts A and B by
M.I. that the statement holds
true for all positive integers n .

(12)

$$\begin{aligned} \text{Q10 a) } \lim_{x \rightarrow 0} \frac{x}{\tan 3x} &= \frac{1}{3} \lim_{x \rightarrow 0} \frac{3x}{\tan 3x} \quad \checkmark \quad \text{must have workig.} \\ &= \frac{1}{3} \times 1 \\ &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \text{b) } x^2 &= 12y \quad (6p, 3p^2) \\ x &= 6p \\ \frac{dx}{dp} &= 6 \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{dp} \div \frac{dx}{dp} \\ &= 6p \times \frac{1}{6} \end{aligned}$$

$$\begin{aligned} y &= 3p^2 \\ \frac{dy}{dp} &= 6p \end{aligned}$$

$$\begin{aligned} &= p \\ \text{gradient of the normal} &= -\frac{1}{p} \quad \checkmark \end{aligned}$$

$$y - 3p^2 = -\frac{1}{p} (x - 6p)$$

$$-py + 3p^3 = x - 6p$$

$$x + py - 6p - 3p^3 = 0 \quad \checkmark$$

asked for general form

$$\begin{aligned} \text{c) } \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{dx}{\sqrt{1-2x^2}} &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{dx}{\sqrt{2} \sqrt{\frac{1}{2} - x^2}} \\ &= \frac{1}{\sqrt{2}} \left[\sin^{-1} \sqrt{2} x \right]_{-\frac{1}{2}}^{\frac{1}{2}} \quad \checkmark \\ &= \frac{2}{\sqrt{2}} \left[\sin^{-1} \sqrt{2} x \right]_0^{\frac{1}{2}} \\ &= \frac{2}{\sqrt{2}} \times \frac{\pi}{4} \\ &= \frac{\pi}{2\sqrt{2}} \quad \checkmark \\ &= \frac{\pi\sqrt{2}}{4} \end{aligned}$$

Q 10 d) $3 \tan 2\theta = 2 \tan \theta \quad 0 \leq \theta \leq 2\pi$

$$\frac{3(\tan \theta + \tan \theta)}{1 - \tan^2 \theta} = 2 \tan \theta$$

$$6 \tan \theta = 2 \tan \theta (1 - \tan^2 \theta)$$

$$= 2 \tan \theta - 2 \tan^3 \theta \quad \checkmark$$

$$2 \tan^3 \theta + 4 \tan \theta = 0$$

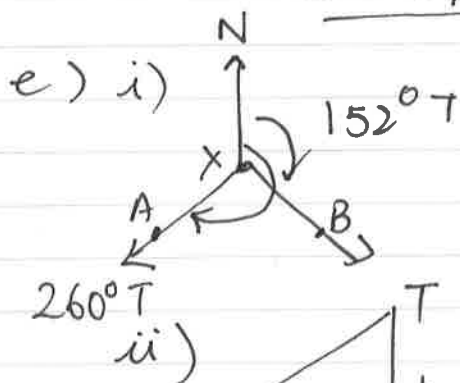
$$2 \tan \theta (\tan^2 \theta + 2) = 0$$

$$\tan \theta = 0$$

$$\theta = 0, \pi, 2\pi \quad \checkmark$$

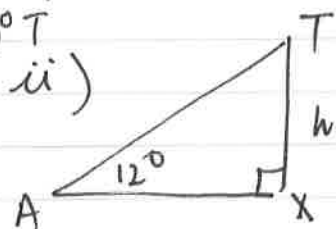
$$\tan^2 \theta = -2 \quad \checkmark$$

no solution



$$\angle A \times B = 260 - 152$$

$$= 108^\circ \quad \checkmark$$

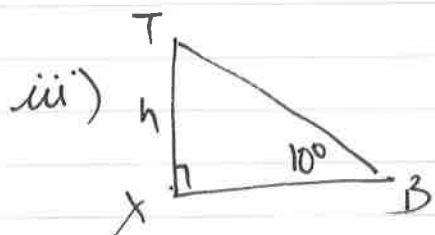


$$\tan 12^\circ = \frac{h}{Ax}$$

$$Ax = \frac{h}{\tan 12^\circ} \quad \checkmark \text{ or similar}$$

$$= h \cot 12^\circ$$

$$= \frac{h}{\tan 78^\circ}$$



$$\tan 12^\circ = \frac{h}{Bx}$$

$$Bx = \frac{h}{\tan 10^\circ}$$

$$= \frac{h}{\tan 80^\circ}$$

$$AB^2 = Ax^2 + Bx^2 - 2Ax \times Bx \times \cos 108 \quad \checkmark$$

$$800^2 = h^2 \tan^2 78 + h^2 \tan^2 80 - 2h^2 \tan 78 \tan 80 \cos 108$$

$$h^2 = \frac{800^2}{\tan^2 78 + \tan^2 80 - 2 \tan 78 \tan 80 \cos 108}$$

$$h^2 = \frac{800^2}{\tan^2 78 + \tan^2 80 - 2 \tan 78 \tan 80 \cos 108}$$

$$h^2 \doteq 9041.218$$

$$h \doteq 95m \quad \checkmark \text{ (nearest metre) } \quad (12)$$

Q11 a) $8 \cos x + 15 \sin x = R \cos(x - \phi)$
 $8 \cos x + 15 \sin x = R \cos x \cos \phi + R \sin x \sin \phi$
 $= \underline{R \cos \phi \cos x} + \underline{R \sin \phi \sin x}$

equating coefficients

$$R \cos \phi = 8$$

$$R \sin \phi = 15$$

$$R = \sqrt{8^2 + 15^2}$$

$$= 17 \checkmark$$

$$\cos \phi = \frac{8}{17}$$

$$\sin \phi = \frac{15}{17}$$

$$\phi = \sin^{-1}\left(\frac{15}{17}\right)$$

$$\hat{=} 61.97^\circ$$

$$\therefore 8 \cos x + 15 \sin x = 17 \cos(x - 62) \checkmark$$

b) LHS = $\frac{\sin \theta - 1 + \cos \theta}{\sin \theta + 1 - \cos \theta}$

$$= \frac{\frac{2t}{1+t^2} - 1 + \frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2} + 1 - \frac{(1-t^2)}{1+t^2}} \checkmark$$

$$= \frac{2t - (1+t^2) + 1-t^2}{1+t^2}$$

$$\frac{2t + (1+t^2) - 1 + t^2}{1+t^2}$$

$$= \frac{2t \cancel{(-1)} - t^2 \cancel{(+1)} + t^2}{1+t^2} \div \frac{2t \cancel{(+1)} + t^2 \cancel{(-1)} + t^2}{1+t^2}$$

$$= \frac{2t - 2t^2}{1+t^2} \times \frac{1+t^2}{2t + 2t^2} \checkmark$$

$$= \frac{2t(1-t)}{1} \times \frac{1}{2t(1+t)}$$

$$= \frac{1-t}{1+t} \quad (t = \tan \frac{\theta}{2})$$

$$= \text{RHS as required}.$$

Q11 c) i) Given $y = px - ap^2$ eq'n of the tgt.
when $y = 0$

$$\begin{aligned} px - ap^2 &= 0 \\ px &= ap^2 \\ \underline{x} &= \underline{ap} \end{aligned}$$

Coordinates of A $(ap, 0)$ ✓

$$\begin{aligned} m_{PA} &= \frac{ap^2}{ap} \\ &= p \end{aligned}$$

$$\begin{aligned} m_{SA} &= \frac{a-0}{0-ap} \\ &= -\frac{1}{p} \end{aligned}$$

$$\begin{aligned} m_{PA} \times m_{SA} &= p \times -\frac{1}{p} \\ &= -1 \end{aligned}$$

∴ $PA \perp SA$ ✓

ii) Since $\angle PAS$ is a right angle
PS is a diameter of a circle
R is the midpoint of PS.
which is the centre C.

$$\begin{aligned} C &\left(\frac{0+2ap}{2}, \frac{a+ap^2}{2} \right) \\ &= C \left(ap, \frac{a(1+p^2)}{2} \right) \end{aligned} \quad \checkmark$$

$$x = ap$$

$$p = \frac{x}{a}$$

$$y = \frac{a \left(1 + \frac{x^2}{a^2} \right)}{2}$$

$$2y = a + \frac{x^2}{a}$$

$$x^2 = a(2y - a) \quad \checkmark$$

Q11 d) i) At $x=0$ and $x=\frac{\pi}{2}$ $\sin^2 x = \sin x$
 For $0 < x < \frac{\pi}{2}$ $0 < \sin x < 1$
 so $\sin^2 x$ will always have a smaller value than $\sin x$. ✓

ii) $V = \pi \int_0^{\frac{\pi}{2}} (y_1^2 - y_2^2) dx$ $y_1 = \sin x$
 $y_2 = \sin^2 x$

$$= \pi \int_0^{\frac{\pi}{2}} (\sin^2 x - \sin^4 x) dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \sin^2 x (1 - \sin^2 x) dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \sin^2 x \cos^2 x dx$$

$$= \frac{\pi}{4} \int_0^{\frac{\pi}{2}} (2 \sin x \cos x)^2 dx$$

$$= \frac{\pi}{4} \int_0^{\frac{\pi}{2}} \sin^2 2x dx$$

$$= \frac{\pi}{4} \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} - \frac{1}{2} \cos 4x \right) dx$$

$$= \frac{\pi}{8} \int_0^{\frac{\pi}{2}} (1 - \cos 4x) dx$$

$$= \frac{\pi}{8} \left[x - \frac{1}{4} \sin 4x \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{8} \left(\left(\frac{\pi}{2} - \frac{1}{4} \sin 2\pi \right) - (0 - 0) \right)$$

$$= \frac{\pi^2}{16} \quad \checkmark$$

(12)